Advanced V&V for motor control

- Motor Control Sensitivity Analysis

Contents

- How does Advanced VV works?
- Structure for Advanced VV
- Harness Block.
- Scripts
- Report
- Folder structure.

Advanced V&V

- Motor control sensitivity analysis.
- Integration testing with Transient Simulation Architecture.
- Verification of algorithms that manage failure reaction.
- Sensitivity analysis on plant parameters variation.
- Automatic Execution of test.
- Robustness of controller.

User Case#1

Description: Evaluate motor controllability under motor parameter variation.

Motor Parameters: Resistance, magnetic flux.

Change parameters [once per time] after motor reaches steady state speed [Test to Fail Mode].

Note-Since resistance and flux cannot be varied directly in Sim Arch, so stator and rotor temperature has been varied.

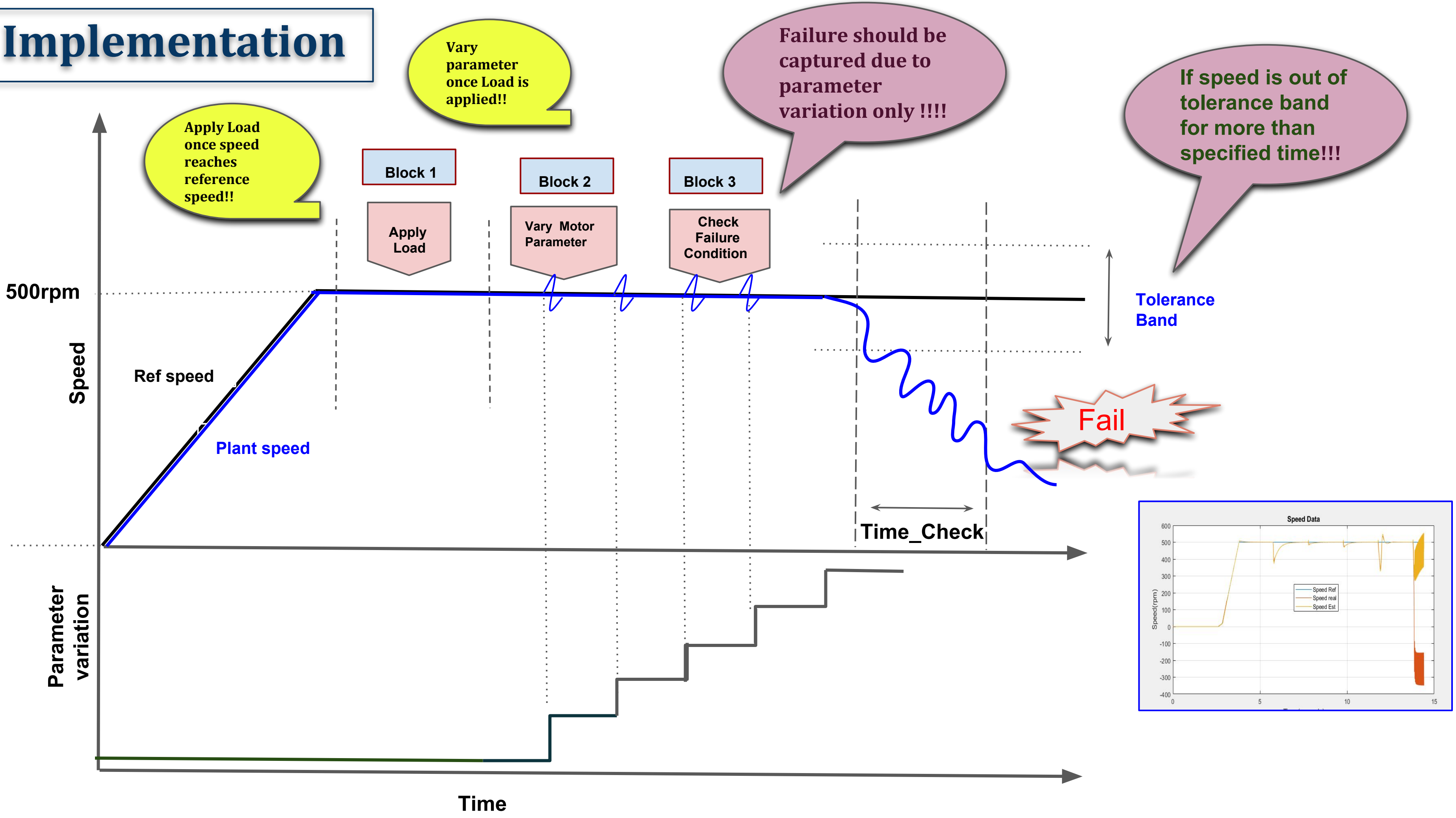
Ranges: Speed\torque matrix [to be applied in dyno load mode].

Speed [Drum]	Torque[Drum]		
50 rpm	0Nm	15Nm	25 Nm
500rpm	0Nm	5Nm	10Nm
1000 rpm	0Nm	3.5Nm	7Nm

Report:

- Rotor position error (Thata Error).
- Torque estimation error (Torque Error).
- Max parameter variation before failure [motor stall condition].

Implementation



Structure

Advanced VV

Scripts

Automatic Script.m

--Allows to run the test with the given speed/torque.

--It manages the output.

Initialization.m

--Allows user to input the Parameters and limits.

--Allows the user for selection of type of load,stream and test.

Harness Model

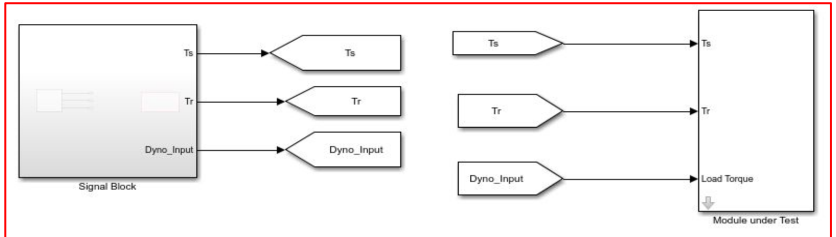
Signal block

It Inputs the parameter variation signal to the module which is under test.

Exported block

Plant

Controller



Library

Library includes detailed documentation of the blocks created in Advanced VV.

Initialization Scripts

```
% Last Modified by v1.1 14-Feb-2019 16:08:20
%---------------------------------------------------------------------------
% Author:      Madhura Patil
% Date:        October 12th, 2018
%---------------------------------------------------------------------------


%---------------------------------------------------------------------------
% Parameter Initialization
%    Description:  This Function is used to load the parameters variable given
%    to simulink Model  used for testing the sBPm by varying  resistance and flux
%    and thus interns temperature of stator and rotor temperature is varied.
%---------------------------------------------------------------------------

%---------------------------------------------------------------------------
% PLEASE NOTE -This script is used in Automatic_script_Temp.m file for calling parameters.
%---------------------------------------------------------------------------

%---------------------------------------------------------------------------
% Edit the below parameters to modify the response.
%---------------------------------------------------------------------------

addpath(genpath(pwd));
%......Input Parametr value.........%
Automatic_Test.Speed=[500;5000;1200];              % .....To enter Speed values(rpm).............
Automatic_Test.Dyno_Load_Vector=[0.5;0.8;1.3];     % .....To enter Dyno Load values(Nm).........
Automatic_Test.Accerelation=500;                   % .....To enter Accerelation of Speed(m/s)....
Automatic_Test.Speed_Tolerance=20;                 % .....To enter Speed Tolerance in(%)........
Automatic_Test.Percentage_Variation=50;            %......To enter the Percentage Variation of th

%...Flag setting for selection of variation of parameter......%

%Uncomment either one of it as per the selection of test,that is by  varying

%Automatic_Test.parameter_var= 'Flux';
 Automatic_Test.parameter_var= 'Rs';



%.......Demagnetisation value........%
Automatic_Test.dm=-0.0021;


%......Input Parametr value to Matlab Functions..........%

%......Status Function.......................%
Automatic_Test.Status_Time_Check=2;             %...Status time check in (sec)
Automatic_Test.Status_Tolerance=0.05;           %...Status Tolerance time in (

%......Dyno Input Function...................%
Automatic_Test.Dyno_Step_Size=0.001;            %...Dyno Step Size value in (s

%......Parameter Input Function..............%
Automatic_Test.Para_time_check=2;               %...Generation of Prameter inp
```

Instruction to use m file

Enter/modify the motor parameters such as speed acc load etc

Enter/modify the controller parameters such as speed tolerance,% variation etc

Selection of test. For e.g by varying resistance /flux.

Enter/modify other parameters as time checks for step size,failure time etc.

Automatic Scripts

```
% Last Modified by v1.1 14-Feb-2019 16:08:20
%--------------------------------------------------------------------------
% Author:       Madhura Patil
% Date:         October 12th, 2018
%--------------------------------------------------------------------------

%--------------------------------------------------------------------------
% Description:  This Script is used to run and test performance of motor
% against defined speed and load torque.
% It checks the robusteness of the motor under test by generating the failure
% values of the tests.
%--------------------------------------------------------------------------

%--------------------------------------------------------------------------
% Requirements to run the test
%   Electrical Requirement:- Permanent Magnet Motor
%   Mechanical Requirement:- Dyno Load
%--------------------------------------------------------------------------

%--------------------------------------------------------------------------
% Please Note- Before running this script, Parameter_Initialization file has
%               to be updated to edit the parameters required to run the test
%--------------------------------------------------------------------------
```

Instructions

```
%.......Opening the model model..........%
addpath('..\..\..\advanced_V_V\tests\parameter_variation\harness_model')
open('Harnes_Temp_Vary.slx');

Parameter_Initialization;
```

Open harness model and call initialization file.

```
%.........To  create and name the folder by current date and time.......%
dname = uigetdir('C:\');
currDate = strrep(datestr(datetime), ':', '_');
bc=strcat(dname,'\',currDate);
```

Create folder to save figure and .csv file.

```
%.......Parameter Intialization for the selection of Ts and Tr signalInput..
h1 = Simulink.findBlocks(gcs,'Name','Parameter Value');
h2 = Simulink.findBlocks(gcs,'Name','Temp Input');
%.......Parameter Intialization for the selection of Parameter (RS) signalIn
if (strcmp(Automatic_Test.parameter_var,'Rs'))
set_param(h1,'init_value','Motor.Electrical.Rs_a_Ref');
set_param(h2,'Init_Value','Motor.Electrical.Rs_a_Ref');
set_param(h2,'Temp_Ref','Motor.Mechanical.Rs_Temperature_Ref');
set_param(h2,'alpha','Motor.Electrical.alpha_s');
elseif(strcmp(Automatic_Test.parameter_var,'Flux'))
%.......Parameter Intialization for the selection of Parameter (Flux) signal
set_param(h1,'init_value','Motor.Electrical.PM_Flux_f01');
set_param(h2,'Init_Value','Motor.Electrical.PM_Flux_f01');
set_param(h2,'Temp_Ref','Motor.Mechanical.PM_Temperature_Ref');
set_param(h2,'alpha','Automatic_Test.dm');
else
  msg='Please select type of variation either Rs or Flux in Automatic_Test.p
 error(msg)
end
```

Masking parameters as per selection of file.

```
%..........Creates an empty table to collect results....%
Results = table([],[],[],[],[],[],[],[],[], ...
'VariableNames',{'Speed_ref','Torque_ref', Automatic_Test.parameter_var,'The
    'Theta_error_std', 'Torque_error_mean', 'Torque_error_std', 'Speed_mean'
```

Create a Report format.

```
%......Iteration.....%

for i= 1:length(Automatic_Test.Speed)
    Automatic_Test.Speed_Ref=Automatic_Test.Speed(i);
     for j=1:length(Automatic_Test.Dyno_Load_Vector)
          Automatic_Test.Dyno_Load=(Automatic_Test.Dyno_Load_Vector(j));
```

Create loop to run the test against Speed and Torque/Parameter matrix.

```
        %.....Simulate the Model...................%
        sim('Harnes_Temp_Vary.slx');
```

Simulate the Harness Model.

```
         %..... To collect statistics............ %
         if (strcmp(Automatic_Test.parameter_var,'Rs'))
         interval_index = get_intervals(Ts.signals.values');
        elseif(strcmp(Automatic_Test.parameter_var,'Flux'))
```

To collect statistics such as mean/standard deviation of Theta and Torque error.

```
%.......To collect the parameters values...........%
 Speed_ref_prm=Automatic_Test.Speed_Ref *ones(size(Theta_Error_mean_values));   %..
 Dyno_ref_prm = Automatic_Test.Dyno_Load * ones(size(Theta_Error_mean_values)); %..
```

To collect values of varied motor parameters at each step interval.

```
%........To get the data of parameter variation.....%
if (strcmp(Automatic_Test.parameter_var,'Rs'))
step = Automatic_Test.Percentage_Variation*Motor.Electrical.Rs_a_Ref/100;
Parameter_Var = (Motor.Electrical.Rs_a_Ref:step:(Motor.Electrical.Rs_a_Ref+step*(le

elseif(strcmp(Automatic_Test.parameter_var,'Flux'))

step = Automatic_Test.Percentage_Variation*Motor.Electrical.PM_Flux_f01/100;
Parameter_Var = (Motor.Electrical.PM_Flux_f01:step:(Motor.Electrical.PM_Flux_f01+st
end
```

To plot the Report.

```
  %.........To update the results table..........%
  Results = [Results ; table(Speed_ref_prm',Dyno_ref_prm',Parameter_Va
       Theta_Error_std_values',Torque_Error_mean_values',Torque_Error_
       Speed_Plant_mean_values',Speed_Plant_std_values', ...
  'VariableNames',{'Speed_ref','Torque_ref', Automatic_Test.parameter_
       'Theta_error_std', 'Torque_error_mean', 'Torque_error_std', 'Spe

  %......Naming the figure file against speed and torque....%
 r=strcat('Speed_',num2str(Automatic_Test.Speed(i)),'_Load_',strrep(n
```

Naming the figure against its speed and torque.

```
  %..... To Plot the Speed, torque and theta error..........%
  y = figure('Name','Run','NumberTitle','off');
  subplot(2,2,1);
  plot(Speed);
  ylabel('Speed(rpm)')
  title('Speed Data')
  legend('Speed Ref','Speed real','Speed Est')
  grid on;
  subplot(2,2,2);
```

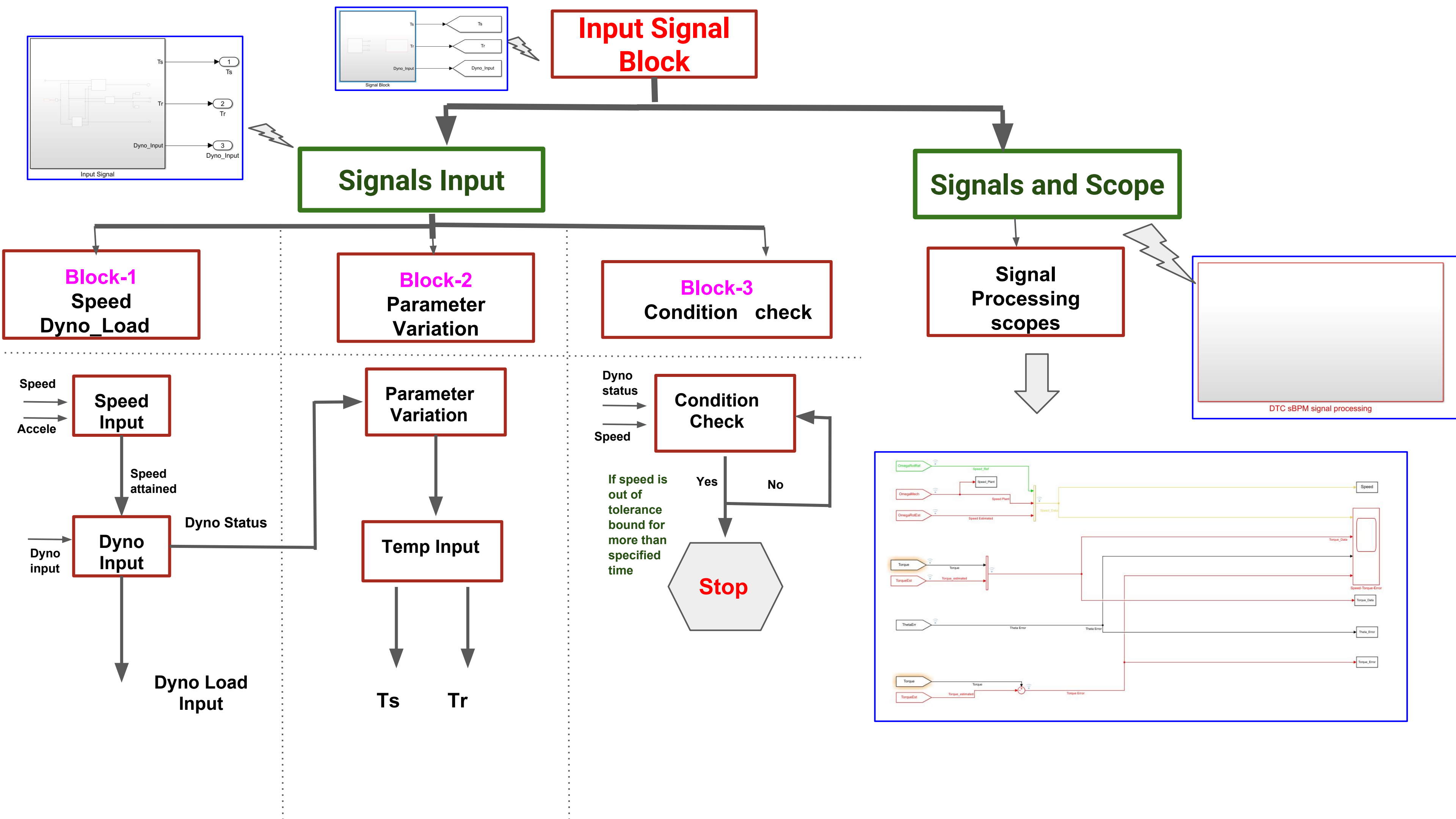
Plot and format each figure by giving label,title,legends etc.

```
function [interval_index] = get_intervals(parameter_signal)
    parameter_signal_shifted = [parameter_signal(1) parameter_signal];
    parameter_signal = [parameter_signal parameter_signal(end)];
    index_signal = parameter_signal_shifted - parameter_signal;

    k=find(index_signal~=0);
    if k
        %.....Adding first interval with nominal value.....%
        interval_index = [k(1)-(k(2)-k(1)) k];
    else
        interval_index = [];
    end
end
```

To save the Report and .fig file in created folder.

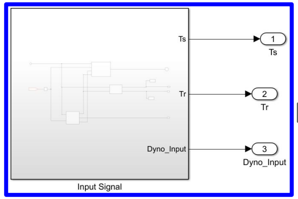
```
function [mean_values,std_values] = statistics(signal, interval_index)
    n=length(interval_index)-1;
    mean_values = zeros(1,n);
    std_values = zeros(1,n);
    for i=1:n
        interval=interval_index(i+1)-interval_index(i);  % To get the interv

        start_1=interval_index(i)+floor(interval*0.2);   % To get the data a
        final=interval_index(i+1)-floor(interval*0.01);  % To get the data be

        windowed_value=signal(start_1:final);
        mean_values(i)=mean(windowed_value);
        std_values(i)=std(windowed_value);
    end
end
```

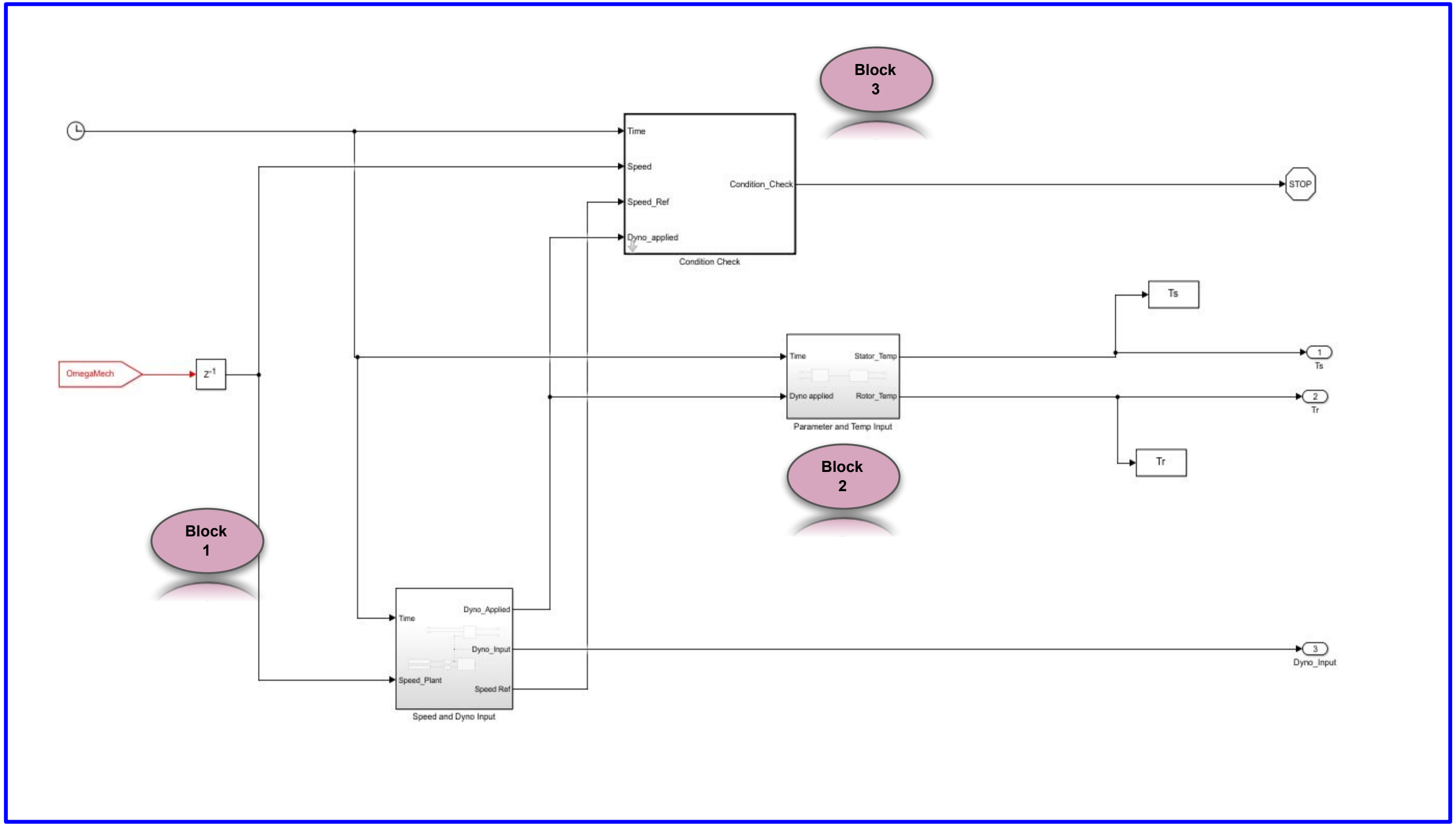
Function to calculate statistics (mean and standard deviation) of each signal at each interval.



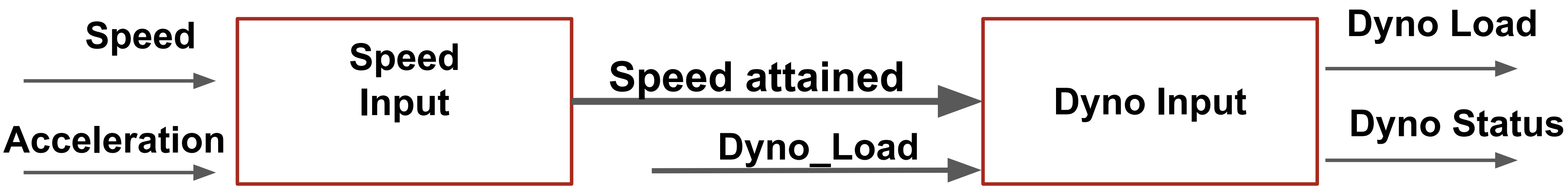
Input Signal Block
Ts
Tr
Dyno_Input
Signal Block
Ts
Tr
Dyno_Input
Input Signal
Signals Input
Signals and Scope
Block-1
Speed
Dyno_Load
Block-2
Parameter Variation
Block-3
Condition check
Signal Processing scopes
DTC sBPM signal processing
Speed
Accele
Speed Input
Speed attained
Dyno Status
Dyno input
Dyno Input
Dyno Load Input
Parameter Variation
Temp Input
Ts
Tr
Dyno status
Speed
Condition Check
Yes
No
If speed is out of tolerance bound for more than specified time
Stop

Matlab Blocks of Signal Input

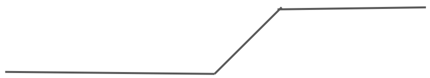




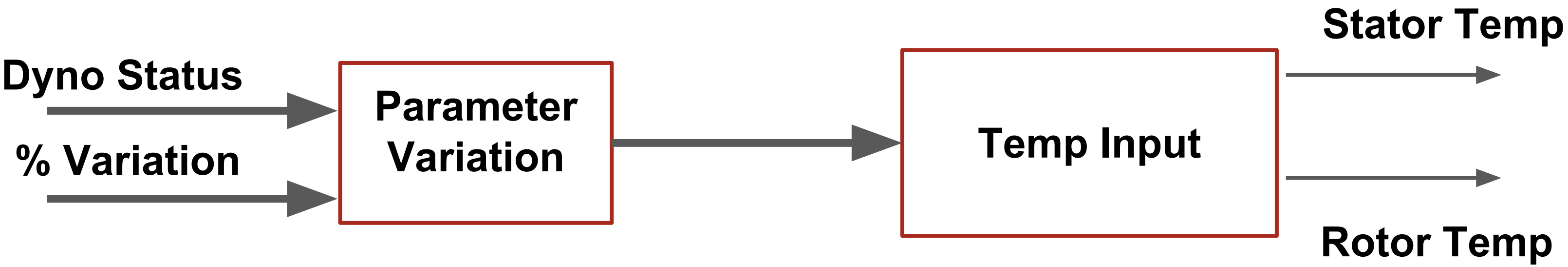
Block-1



- **Once the plant speed reaches reference speed status will be sent to Dyno input to apply Dyno Load.**
- **Dyno Load will be applied once status is 1.**
- **Dyno Load is an Ramp+step input signal.**



Block-2

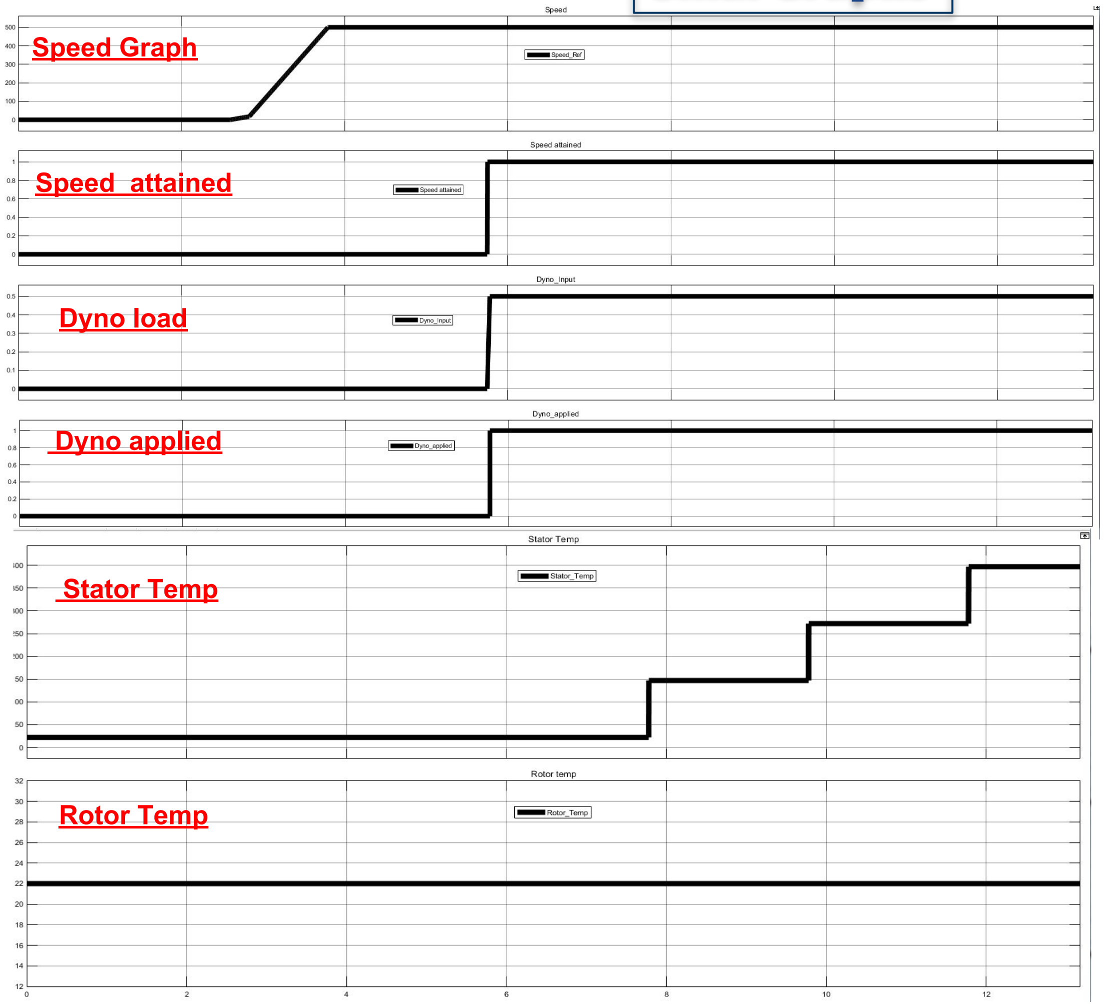


- **Once the Dyno status is 1, after certain time check, parameter value will generate step input.**
- **Corresponding to parameter that is changed, temperature input is generated.**

Block-3

- Condition will check once dyno load applied.
- Speed is within tolerance band.
- If speed is out of tolerance bound for more than specified time, simulation will stop.

Sinal Graphs



Speed Graph

When plant speed reaches ref speed, Speed attained Status is high.

Speed attained Status is high Dyno load has applied.

When Dyno load has applied, Dyno applied status is high.

Once Dyno applied status is high, parameter variation begins.

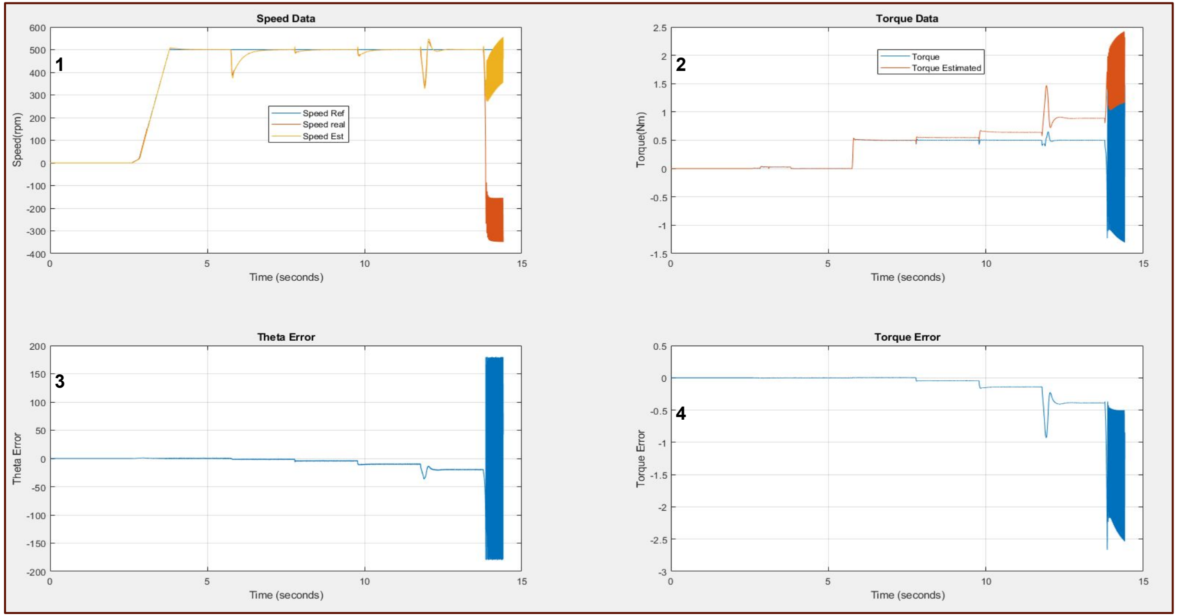
Vary one parameter at a time.

Other parameter retains its nominal value during the test.

Results

Format of figure

1. **Speed-Reference,Plant and Controller speed**
2. **Torque- Plant and Controller torque.**
3. **Theta Error**
4. **Torque error**



Format of report

https://docs.google.com/spreadsheets/d/1OggEb_Yh8j9XC7TgvE7_nNsHmTqi0iXSu4GJ6xzH1ZA/edit#gid=853431869

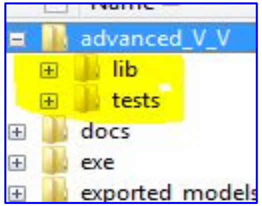
Report

Speed_ref	Torque_ref	Rs	Theta_error_mean	Theta_error_std	Torque_error_mean	Torque_error_std	Speed_mean	Speed_std
50	5	2.87	1.10	0.02	0.00	0.00	582.31	3.79
50	5	4.305	-0.55	0.02	-0.03	0.00	584.71	0.42
50	5	5.74	-3.95	0.02	-0.07	0.00	584.57	0.65
50	5	7.175	-9.44	0.05	-0.16	0.00	584.35	1.13
50	10	2.87	1.20	0.02	0.00	0.00	581.49	4.95
50	10	4.305	-2.04	0.03	-0.11	0.00	583.94	1.59
50	10	5.74	-7.90	0.06	-0.27	0.00	583.90	1.96
50	15	2.87	1.31	0.02	0.01	0.00	581.25	5.30
50	15	4.305	-3.38	0.06	-0.24	0.00	583.06	3.18

Folder structure

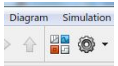
MCU_Sim_Arch

advanced_V_V

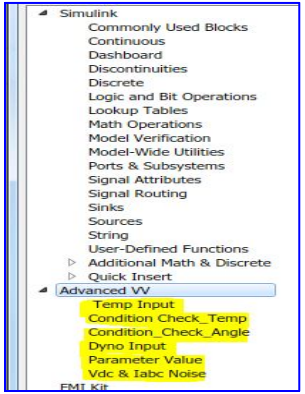


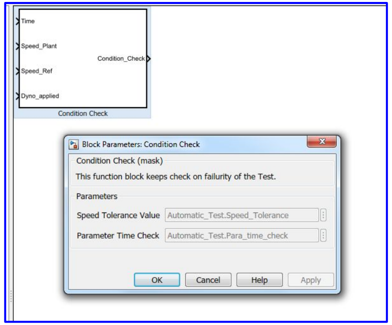
lib (Library)

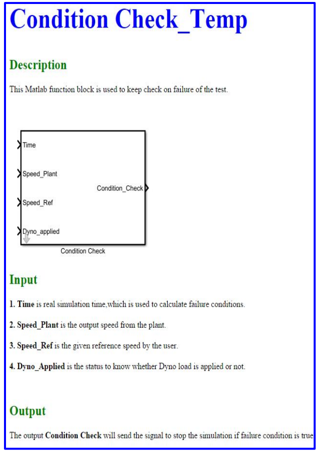
This folder contains library of each matlab blocks created for advanced VV testing purpose.



Go to library browser







Condition Check_Temp

Description

This Matlab function block is used to keep check on failure of the test.

Input

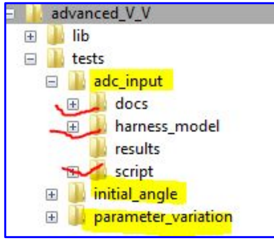
1. **Time** is real simulation time which is used to calculate failure conditions.
2. **Speed_Plant** is the output speed from the plant.
3. **Speed_Ref** is the given reference speed by the user.
4. **Dyno_Applied** is the status to know whether Dyno load is applied or not.

Output

The output **Condition Check** will send the signal to stop the simulation if failure condition is true

tests

--adc_input
--initial-Angle
--Parameter variation



Docs-Help document/guidelines to conduct tes.

Harness_model- Contains harness model as .slx file.

Results - Folder to save the results.

Script - Contains two script, Automatic and Initialization.m files